

Guideline for the ideal answer on the chart critique about preventive maintenance

Key message

- Name closure rate (%) as the single metric leadership needs — absolute order counts are secondary.
- Explicitly identify the dual-axis structure (bars + line + labels) as the source of overload.

Concrete redesign

- Remove open bars entirely (or grey them out) — leadership does not need to see open orders.
- Either keep only the % line (simplest) or separate into two charts: one bar chart for counts, one line/bar chart for closure rate.
- Remove or simplify the right Y-axis (% labels are redundant if shown on the line).
- Use descending order if showing a single bar chart so the ranking is immediately visible.

Audience-centred justification

- Leadership needs to compare closure rates across asset types at a glance — every design choice should serve that goal.
- Removing counts is not a loss of information; it is a gain in focus.

Guideline for the ideal answer on the chart critique about ARR

The three main issues

- Label clutter on line charts — every MOB data point is labelled, making the trend unreadable. Fix: add Y-axis, remove inline labels (keep only first/last or key inflection).
- Commentary box duplicates chart content and creates cognitive overload. Fix: reduce to 2–3 bullets or replace with action-oriented chart titles / KPI cards.
- Bar charts lack a shared axis — lie factor distorts the MP vs SP comparison. Fix: use a 100% stacked bar or align axes across panels.
- Missing legend and undefined abbreviations (MOB, AE, PLG, SP, ARR). Fix: add legend, spell out terms.
- No visual hierarchy — title, subtitle, charts, and commentary all compete equally. Fix: differentiate with size, weight, and colour contrast.

For full marks each issue must

- Name at least one design principle (cognitive load, data-ink ratio, or audience-centred design) to explain *why* it is a problem.
- Propose a specific fix — not just "make it simpler" but *how*.

Main issues across the cohort

Several recurring weaknesses emerged across both questions:

- **Jumping to solutions without diagnosing the problem first.** The most widespread issue was proposing a new chart type without first explaining what makes the original cluttered. The redesign needs to follow from a diagnosis — not replace it. Many students wrote "I would use a bar chart" without ever stating why the existing chart fails.

- **Not naming design principles.** Very few students explicitly invoked the three core concepts the question was built around: cognitive load, data-ink ratio, and audience-centred design. Observations tended to stay at a functional level ("it's hard to read", "there's too much information") rather than being grounded in theoretical reasoning.
- **Counterproductive redesigns in the preventive maintenance chart.** A significant number of students proposed stacked bar charts, pie charts, or scatter plots as improvements to the preventive maintenance chart. All three of these alternatives actually make closure rate comparison harder, not easier. A stacked bar obscures the closed proportion by shifting its baseline; a pie chart makes cross-category comparison nearly impossible; a scatter plot has no meaningful application here. The better answer was to simplify toward the single metric leadership cares about — the closure rate percentage — rather than reorganise the complexity.
- **Fewer than three distinct issues identified in the ARR chart.** The question explicitly asked for at least three major issues. Several students identified only one or two, or listed problems that were too similar to count as distinct. A missing Y-axis and a missing legend are two separate issues; commentary overload and poor storytelling, while related, need to be treated as separate arguments with separate fixes to earn full marks.
- **Vague improvement proposals.** Fixes such as "make it simpler", "use better colours", or "reduce the text" appeared frequently and were not creditworthy. A good improvement proposal names the specific element being changed, describes the change precisely, and explains how it serves the audience. For example: "remove inline data labels from the line charts and add a Y-axis instead, so the audience can read the trend without processing 24 individual numbers" is a specific, justified fix.
- **Missing audience-centred justification.** Many redesigns were justified on aesthetic grounds — "more visually appealing", "cleaner", "easier on the eye" — rather than being connected to what the audience actually needs to do with the information. In the preventive maintenance chart, leadership needs to compare closure rates across four asset types at a glance. Every design decision should be traced back to that goal. Removing open bars is not a loss of information; it is a gain in focus for that specific audience.
- **Misreading the ARR dashboard.** Several students misidentified or mischaracterised issues — for example, attributing the data-ink problem to "too much white space" (when the real issue is label density), or flagging line overlap rather than label clutter on the MOB charts. This suggests some students read the dashboard quickly rather than analytically. Good visual critique requires slowing down and asking: what is each element doing, and is it earning its place?
- **Completeness mindset working against audience-centred design.** A notable minority of students argued that their redesign was better because the audience would "not lose any information". This reflects a completeness mindset that directly contradicts what audience-centred design teaches: the goal is not to show everything, but to show exactly what the audience needs to make a decision — and nothing more.

Finally, the case study was drawn from this blog: [different takeaway, different graph](#)